

Exercise 0: Recap of Diff Geo

Recall that for vector fields $X, Y, Z \in \mathfrak{X}(M)$ and smooth functions $f, g \in C^\infty(M)$, the connection ∇ satisfies:

1. $\nabla_f X + gY = f\nabla_X Z + g\nabla_Y Z$,
2. $\nabla_X(Y + Z) = \nabla_X Y + \nabla_X Z$,
3. $\nabla_X(fY) = (\nabla_X f)Y + f\nabla_X Y$, with $\nabla_X f = X(f)$.

Let x^μ be a local coordinate system, with associated coordinate basis $\{\partial_\mu\}$. We define the Christoffel symbols as $\nabla_{\partial_\mu} \partial_\nu = \Gamma_{\mu\nu}^\rho \partial_\rho$. Now let $V = V^\mu \partial_\mu$ and $W = W^\nu \partial_\nu$ be two arbitrary vector fields, and let $\omega = \omega_\mu dx^\mu$ be a covector field.

a) Using the axioms and definitions above, prove

$$\nabla_V W = V^\mu (\partial_\mu W^\rho + \Gamma_{\mu\nu}^\rho W^\nu) \partial_\rho. \quad (1)$$

Then, recalling that $\omega(V) = \omega_\mu V^\mu$, prove that

$$\nabla_V \omega = V^\mu (\partial_\mu \omega_\nu - \Gamma_{\mu\rho}^\nu \omega_\rho) dx^\nu. \quad (2)$$

Identify the expressions in brackets.

$$\begin{aligned} \nabla_V W &= \nabla_V (W^\mu \partial_\mu) \\ &= (\nabla_V W^\mu) \partial_\mu + W^\mu \nabla_V \partial_\mu \\ &= V(W^\mu) \partial_\mu + W^\mu \nabla_V \partial_\mu \\ &= [V^\nu \partial_\nu W^\mu] \partial_\mu + W^\mu \nabla_{V^\nu \partial_\nu} \partial_\mu \\ &= [V^\nu \partial_\nu W^\mu] \partial_\mu + W^\mu V^\nu \nabla_{\partial_\nu} \partial_\mu \\ &= [V^\nu \partial_\nu W^\mu] \partial_\mu + W^\mu V^\nu \Gamma_{\nu\mu}^\rho \partial_\rho \\ &= V^\nu [\partial_\nu W^\mu + W^\mu \Gamma_{\nu\mu}^\rho] \partial_\rho. \end{aligned}$$

Then, using Leibniz,

$$\nabla_V (\omega(Y)) = (\nabla_V \omega)(Y) + \omega(\nabla_V Y)$$

$$(\nabla_V \omega)(Y) = \nabla_V (\omega(Y)) - \omega(\nabla_V Y)$$

$$= V^\mu \partial_\mu (\omega_\nu Y^\nu) - \omega_\mu (\nabla_V Y)^\mu$$

$$= V^\mu (\partial_\mu \omega_\nu Y^\nu + \omega_\nu \cancel{\partial_\mu Y^\nu}) - \omega_\mu V^\rho (\cancel{\partial_\nu Y^\rho} + \Gamma_{\nu\rho}^\sigma Y^\sigma)$$

$$= V^\mu (\partial_\mu \omega_\nu Y^\nu - \omega_\rho V^\rho \Gamma_{\mu\nu}^\rho Y^\nu)$$

$$= V^\mu [\partial_\mu \omega_\nu - \omega_\rho \Gamma_{\mu\nu}^\rho] Y^\nu$$

Since this holds for all Y ,

$$\nabla_V \omega = V^\mu (\partial_\mu \omega_\nu - \Gamma_{\mu\nu}^\rho \omega_\rho) dx^\nu$$

b) Using the identity $\Gamma_{\mu\nu}^\mu = \partial_\nu \ln \sqrt{|g|}$, where $g = \det(g_{\mu\nu})$, show that

$$\nabla_\mu V^\mu = \frac{1}{\sqrt{|g|}} \partial_\mu (\sqrt{|g|} V^\mu). \quad (3)$$

$$\nabla_\mu V^\mu = \partial_\mu V^\mu + \Gamma_{\mu\nu}^\mu V^\nu$$

$$= \partial_\mu V^\mu + (\partial_\nu \ln \sqrt{|g|}) V^\nu$$

$$= \partial_\mu V^\mu + \frac{1}{\sqrt{|g|}} (\partial_\nu \sqrt{|g|}) V^\nu$$

$$= \frac{1}{\sqrt{|g|}} [\sqrt{|g|} \partial_\mu V^\mu + (\partial_\nu \sqrt{|g|}) V^\nu]$$

$$= \frac{1}{\sqrt{|g|}} \partial_\nu (\sqrt{|g|} V^\nu)$$

A tangent vector V^a is *causal* if it is timelike or lightlike, that is, if $V_a V^a \leq 0$. Then, we say that a spacetime M is *time-orientable* if it admits a *time-orientation*: an everywhere causal vector field T^a . At any $p \in M$, the metric determines two light-cones in $T_p M$. We can then define the light-cone in which T^a lies as the "future" light-cone, and the other one as the "past" light-cone. Then, we say that any causal vector X^a is *future-directed* if it lies in the same light-cone as T^a , and *past-directed* otherwise.

a) Let X^a and Y^a be two causal vectors at a point $p \in M$. Show that

$$X_a Y^a \leq 0 \iff X^a \text{ and } Y^a \text{ lie in the same light-cone.} \quad (1)$$

First we pick normal coordinates such that $g_p = \text{diag}(-1, 1, 1, 1)$.

$$X^a Y_a = -X^0 Y^0 + \sum_i x^i y^i$$

We know that $(x^0)^2 \geq (x^1)^2 + (x^2)^2 + (x^3)^2$ and similar for y .

$$\text{Using Cauchy-Schwarz, } (\sum_i x^i y^i)^2 \leq (\sum_i (x^i)^2) \cdot (\sum_i (y^i)^2)$$

If X and Y are in the same light cone, $X^0 Y^0 \geq 0$ and hence

$$\begin{aligned} -X^0 Y^0 + \sum_i x^i y^i &\leq -X^0 Y^0 + \left| \sum_i x^i y^i \right| \\ &\leq -X^0 Y^0 + \sqrt{\sum_i (x^i)^2} \sqrt{\sum_i (y^i)^2} \\ &\leq -X^0 Y^0 + |X^0| |Y^0| \quad (\text{timelike}) \\ &\leq 0 \end{aligned}$$

Conversely, suppose that X and Y are in different light cones. By a similar argument, $X^0 Y^0 \leq 0$ and

$$\begin{aligned} -X^0 Y^0 + \sum_i x^i y^i &= |X^0| |Y^0| + \sum_i x^i y^i \\ &\geq |X^0| |Y^0| - \left| \sum_i x^i y^i \right| \\ &\geq |X^0| |Y^0| \end{aligned}$$

The action for a minimally-coupled scalar field ϕ is

$$S_M = - \int_M d^4x \sqrt{-g} \left[\frac{1}{2} g_{\mu\nu} \partial^\mu \phi \partial^\nu \phi + V(\phi) \right], \quad (2)$$

where $V(\phi)$ is the potential of the scalar field.

a)* Imposing Dirichlet boundary conditions, $\delta\phi|_{\partial M} = 0$, vary S_M with respect to ϕ to find

$$\square\phi - V'(\phi) = 0. \quad (3)$$

Do not use the Euler-Lagrange equations, show explicitly how to deal with the boundary term.

b)* Vary S_M with respect to the inverse metric $g^{\mu\nu}$ to find the stress-energy tensor,

$$T_{\mu\nu} = \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \nabla^\rho \phi \nabla_\rho \phi - g_{\mu\nu} V(\phi). \quad (4)$$

a)

$$\begin{aligned} \delta S_M &= - \int d^4x \sqrt{-g} \left[\frac{1}{2} (\partial_\mu (\delta\phi) \partial^\mu \phi + \partial_\mu \phi \partial^\mu (\delta\phi)) + V'(\phi) \delta\phi \right] \\ &= - \int d^4x \sqrt{-g} \left\{ \frac{1}{2} \left[\underbrace{\partial_\mu (\delta\phi \partial^\mu \phi)}_{\text{boundary terms}} - \delta\phi \partial_\mu \partial^\mu \phi + \partial^\mu (\delta\phi \partial_\mu \phi) - \delta\phi \partial^\mu \partial_\mu \phi \right] + V'(\phi) \delta\phi \right\} \end{aligned}$$

Since ϕ is a scalar, $\partial_\mu \phi = \nabla_\mu \phi$. Thus,

$$\nabla_\mu (\delta\phi \nabla^\mu \phi) = (\nabla_\mu A)^\mu$$

where $A^\mu = \delta\phi \nabla^\mu \phi$ (this is true because derivatives can be commuted). Thus we can integrate by parts

$$\begin{aligned} \int_M d^4x \sqrt{-g} \nabla_\mu A^\mu &= \int_{\partial M} d^3x \sqrt{-g} n_\mu A^\mu \\ &= \int_{\partial M} d^3x \sqrt{-g} (n_\mu \nabla^\mu \phi) \delta\phi \stackrel{0}{=} 0 \end{aligned}$$

Thus all boundary terms vanish by the divergence theorem.

$$\begin{aligned} \delta S_M &= \int d^4x \sqrt{-g} \left\{ \frac{1}{2} (\delta\phi \partial_\mu \partial^\mu \phi + \delta\phi \partial^\mu \partial_\mu \phi) - V'(\phi) \delta\phi \right\} \\ &= \int d^4x \sqrt{-g} \left\{ \partial_\mu \partial^\mu \phi - V'(\phi) \right\} \delta\phi \end{aligned}$$

Since this holds for all $\delta\phi$,

$$\begin{aligned} &\partial_\mu \partial^\mu \phi - V'(\phi) \\ &= \nabla_\mu \nabla^\mu \phi - V'(\phi) \\ &= \square\phi - V'(\phi) = 0 \end{aligned}$$

b)* Vary S_M with respect to the inverse metric $g^{\mu\nu}$ to find the stress-energy tensor,

$$T_{\mu\nu} = \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \nabla^\rho \phi \nabla_\rho \phi - g_{\mu\nu} V(\phi). \quad (7)$$

lemma (from Tong): $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}$

$$\begin{aligned} S_M &= - \int d^4x \sqrt{-g} \left[\frac{1}{2} \partial_\mu \phi \partial^\mu \phi + V(\phi) \right] \\ &= - \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + V(\phi) \right] \\ \delta S_M &= - \int d^4x \left\{ -\frac{1}{2} \sqrt{-g} g_{\alpha\beta} \delta g^{\alpha\beta} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + V(\phi) \right] + \sqrt{-g} \left[\frac{1}{2} \delta g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right] \right\} \\ &= - \int d^4x \sqrt{-g} \left\{ -\frac{1}{2} g_{\alpha\beta} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + V(\phi) \right] + \frac{1}{2} \partial_\alpha \phi \partial_\beta \phi \right\} \delta g^{\alpha\beta} \end{aligned}$$

Since $T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}$, we have

$$\begin{aligned} T_{\mu\nu} &= \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} g^{\sigma\tau} \partial_\sigma \phi \partial_\tau \phi - g_{\mu\nu} V(\phi) \\ &= \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \nabla_\rho \phi \nabla^\rho \phi - g_{\mu\nu} V(\phi) \end{aligned}$$

because $\partial_\nu \phi = \nabla_\nu \phi$ for scalar ϕ .

Exercise 3: Cosmology

The flat FLRW metric, describing a homogeneous and isotropic flat Universe, reads

$$ds^2 = -dt^2 + a^2(t) (dx^2 + dy^2 + dz^2) . \quad (8)$$

Notice that the x, y, z coordinates do not measure *physical* distances. They are called *comoving coordinates*, since they “move” with the expansion of the Universe.

a)* Derive the geodesic equation for particles in a FLRW universe after arguing that we can focus solely on motion along the x direction without loss of generality. Argue that the energy E of massless particles decays with the expansion of the Universe. Comment on the physical relevance of this result.

Hint: Recall that E is the 0th component of the 4-momentum, so $E \propto dt/d\sigma$ with σ being an affine parameter along the geodesic.

By isotropy we can rotate any motion into motion along the x axis

Hence we can pretend that spacetime is 1+1d. We can write the functional

$$L = \int \sqrt{\left(\frac{dx^0}{d\lambda}\right)^2 - a(t)^2 \left(\frac{dx^1}{d\lambda}\right)^2} d\lambda$$

Instead, we work with

$$S = \int d\tau \left[-(\dot{x}^0)^2 + a(t)^2 (\dot{x}^1)^2 \right] d\tau, \quad \dot{x}^\mu := \frac{dx^\mu}{d\tau}$$

$$\begin{aligned} \delta S &= \int d\tau \left[-(\dot{x}^0) \delta \dot{x}^0 + a(t) a'(t) (\dot{x}^1)^2 \delta x^0 + a(t)^2 (\dot{x}^1) \delta \dot{x}^1 \right] \\ &= \int d\tau \left[\ddot{x}^0 \delta x^0 + a(t) a'(t) (\dot{x}^1)^2 \delta x^0 - \frac{d}{d\tau} (a(t)^2 \dot{x}^1) \delta x^1 \right] \end{aligned}$$

Hence the equations of motion are

$$\begin{aligned} \ddot{x}^0 + a(t) a'(t) (\dot{x}^1)^2 &= 0 \\ 0 &= \frac{d}{d\tau} (a(t)^2 \dot{x}^1) \\ &= a(t)^2 \ddot{x}^1 + 2a(t) a'(t) \dot{x}^1 \dot{x}^0 \\ \ddot{x}^1 + 2 \frac{a'(t)}{a(t)} \dot{x}^1 \dot{x}^0 &= 0 \end{aligned}$$

These equations must be understood together with the speed constraint $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -c^2$ (massive) / $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0$ (massless)

Null condition: $-(\dot{x}^0)^2 + a(t)^2 (\dot{x}^1)^2 = 0$

$$\dot{x}^0 = a(t) \dot{x}^1 \quad (\text{wlog } +x \text{ direction})$$

Using this in the conservation law

$$a(t)^2 \dot{x}^1 = \text{const},$$

we find that

$$a(t) \dot{x}^0 = \text{const}$$

so

$$E \propto \dot{x}^0 \propto \frac{1}{a(t)}$$

hence E decreases with time.

b) Read off the Christoffel symbols from the geodesic equation. Then compute the Ricci tensor and the Ricci scalar. You may check your results using `diffgeo`.

Hint: Diagonal metrics that depend only on one coordinate have a diagonal Ricci tensor.

$$\ddot{x}^0 + a(t) a'(t) (\dot{x}^1)^2 = 0$$

$$\ddot{x}^1 + 2 \frac{a'(t)}{a(t)} \dot{x}^1 \dot{x}^0 = 0$$

Geodesic equation

$$\ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0$$

Hence $\Gamma_{11}^0 = a(t) a'(t)$
 $\Gamma_{10}^1 = \frac{a'(t)}{a(t)}$

All other Christoffel symbols are 0.

Riemann Tensor $R^\sigma_{\rho\mu\nu} = \partial_\mu \Gamma_{\nu\rho}^\sigma - \partial_\nu \Gamma_{\mu\rho}^\sigma + \Gamma_{\nu\rho}^\lambda \Gamma_{\mu\lambda}^\sigma - \Gamma_{\mu\rho}^\lambda \Gamma_{\nu\lambda}^\sigma$

Ricci Tensor $R_{\mu\nu} = R^\rho_{\mu\rho\nu}$
 $= \partial_\rho \Gamma_{\nu\mu}^\rho - \partial_\nu \Gamma_{\rho\mu}^\rho + \Gamma_{\nu\mu}^\lambda \Gamma_{\rho\lambda}^\rho - \Gamma_{\rho\mu}^\lambda \Gamma_{\nu\lambda}^\rho$

$R_{11} = \partial_0 \Gamma_{11}^0 - \cancel{\partial_1 \Gamma_{01}^0} + \Gamma_{11}^0 \Gamma_{10}^1 - \cancel{\Gamma_{10}^1 \Gamma_{11}^0}$
all christoffels depend only on x^0

$= \partial_0 [a(t) a'(t)] + \cancel{a(t) a'(t)} \frac{a'(t)}{a(t)}$

$= a''(t) a(t) + 2 a'(t)^2$

$R_{00} = \cancel{\partial_0 \Gamma_{00}^0} - \partial_0 \Gamma_{10}^1 + \cancel{\Gamma_{00}^1 \Gamma_{10}^0} - \Gamma_{01}^1 \Gamma_{01}^0$

$= -\partial_0 \left[\frac{a'(t)}{a(t)} \right] - \left[\frac{a'(t)}{a(t)} \right]^2$

$= -\frac{a''(t)}{a(t)} + \frac{a'(t)^2}{a(t)^2} - \left[\frac{a'(t)}{a(t)} \right]^2$

$= -\frac{a''(t)}{a(t)}$

Note: In 3d we have an extra factor of 3 arising from the spatial sums $\Gamma_{10}^1, \Gamma_{20}^2, \Gamma_{30}^3$! Hence

$R_{00} = -3 \frac{a''(t)}{a(t)}$

Ravi scalar $R = g^{\mu\nu} R_{\mu\nu}$

$$= (-1) \cdot \left(-3 \frac{a''(t)}{a(t)} \right) + 3 \frac{1}{a(t)^2} \left[a''(t) a(t) + 2 a'(t)^2 \right]$$

$$= 6 \left(\frac{a''(t)}{a(t)} + \frac{a'(t)^2}{a(t)^2} \right)$$

Exercise 4: Energy Conditions

a) Prove that DEC $\implies$ WEC.

Weak Energy Condition: For all timelike X , $T_{\mu\nu} X^\mu X^\nu \geq 0$.
 Dominant Energy Condition: For all future directed timelike X ,
 defining $J^\nu = -T^\nu{}_\mu X^\mu$, $J^\nu J_\nu \leq 0$, J is future directed

Proof of DEC $\implies$ WEC: By exercise 1, since X and J are future directed causal, $X_\mu J^\mu \leq 0$
 But $X_\mu J^\mu = -T^\mu{}_\nu X^\mu X^\nu = -T_{\mu\nu} X^\mu X^\nu$
 Hence $T_{\mu\nu} X^\mu X^\nu \geq 0$

b)* The stress-energy tensor of a perfect fluid reads

$$T_{\mu\nu} = (\rho + p) U_\mu U_\nu + p g_{\mu\nu}, \tag{9}$$

where U^μ , a future-directed timelike vector with $U^\mu U_\mu = -1$, is the fluid's 4-velocity, and ρ and p are the fluid's energy density and isotropic pressure, respectively.

Show that

$$\text{NEC} \implies (\rho + p) \geq 0, \tag{10}$$

$$\text{WEC} \implies \rho \geq 0 \quad \text{and} \quad (\rho + p) \geq 0, \tag{11}$$

$$\text{DEC} \implies \rho \geq |p|, \tag{12}$$

$$\text{SEC} \implies (\rho + 3p) \geq 0 \quad \text{and} \quad (\rho + p) \geq 0. \tag{13}$$

• Null Energy Condition: The null energy condition

$$T_{\mu\nu} X^\mu X^\nu \geq 0 \quad \text{for all } X \text{ with } X \cdot X = 0$$

$$T_{\mu\nu} X^\mu X^\nu = (\rho + p) U_\mu X^\mu U_\nu X^\nu + p g_{\mu\nu} X^\mu X^\nu$$

$$= (\rho + p) [U_\mu X^\mu]^2 \geq 0$$

Choose X^μ such that $U_\mu X^\mu$ is arbitrarily small, thus

$$\rho + p \geq 0$$

• Weak Energy Condition: This states that, for any timelike vector field X ,

$$T_{\mu\nu} X^\mu X^\nu \geq 0 \quad \text{for all } X \text{ with } X_\mu X^\mu < 0$$

$$T_{\mu\nu} X^\mu X^\nu = (\rho + p) U_\mu X^\mu U_\nu X^\nu + p g_{\mu\nu} X^\mu X^\nu$$

We pick $X = U$ and find

$$T_{\mu\nu} X^\mu X^\nu = \rho + p - p = \rho \geq 0$$

Since NEC is weaker than WEC $\rho + p \geq 0$

• Dominant Energy Condition: There is also an energy condition which is stronger than the weak condition. For any future-directed timelike vector X , we can define the current

$$J^\mu = -T^{\mu\nu} X_\nu \tag{4.62}$$

This is energy density current as seen by an observer following the lines of X . The dominant energy condition requires that, in addition to the weak energy condition (4.60), the current is either timelike or null, so

$$J_\mu J^\mu \leq 0$$

This is the reasonable statement that energy doesn't flow faster than light.

$$J^\nu = -(\rho + p) U^\mu U^\nu X_\mu - g^{\mu\nu} X_\mu$$

$$J_\mu = -(\rho + p)U_\mu U^\nu X_\nu - pg_{\mu}{}^\nu X_\nu$$

$$\begin{aligned} J_\mu J^\mu &= \left[-(\rho + p)U^\mu U^\nu X_\nu - pg_{\mu}{}^\nu X_\nu \right] \left[-(\rho + p)U_\mu U^\nu X_\nu - pg_{\mu}{}^\nu X_\nu \right] \\ &= (\rho + p)^2 \underbrace{U^\mu U_\mu}_{=-1} (U^\nu X_\nu)^2 + 2\rho(\rho + p)(U^\nu X_\nu)^2 + \rho^2 g^{\mu\nu} X_\mu X_\nu \\ &= (\rho^2 - \rho^2)(U^\nu X_\nu)^2 + \rho^2 X_\mu X^\mu \end{aligned}$$

Now choose X with $X_\mu X^\mu = -1$.

$$\text{Then } J_\mu J^\mu = (\rho^2 - \rho^2)(U^\nu X_\nu)^2 - \rho^2 \leq 0$$

Now we see if $\rho^2 - \rho^2 > 0$, we can take $U^\nu X_\nu \rightarrow \infty$ and violate the DEC.

Thus $\rho^2 - \rho^2 \leq 0$, since $\rho \geq 0$ by WEC, $\rho \geq |p|$

• **Strong Energy Condition:** There is a different, less immediately intuitive, energy condition. This is the requirement that, for any timelike vector field X ,

$$R_{\mu\nu} X^\mu X^\nu \geq 0$$

Using the form of the Einstein equations (4.49), the strong energy condition requires

$$\left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right) X^\mu X^\nu \geq 0$$

for all timelike vector fields X . As before, continuity ensures that we can extend this to timelike and null vector fields, $X \cdot X \leq 0$.

$$T = g^{\mu\nu} T_{\mu\nu}$$

$$= g^{\mu\nu} [(\rho + p)U_\mu U_\nu + pg_{\mu\nu}]$$

$$= -(\rho + p) + 4\rho = -\rho + 3\rho$$

$$\begin{aligned} T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} &= (\rho + p)U_\mu U_\nu + pg_{\mu\nu} - \frac{1}{2}(-\rho + 3\rho)g_{\mu\nu} \\ &= (\rho + p)U_\mu U_\nu + \frac{1}{2}(\rho - p)g_{\mu\nu} \end{aligned}$$

Choose $X = U$. Then

$$\begin{aligned} (T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu}) X^\mu X^\nu &= (\rho + p) - \frac{1}{2}(\rho - p) \\ &= \frac{\rho}{2} + \frac{3\rho}{2} \geq 0 \end{aligned}$$

$$\text{Hence } \rho + 3\rho \geq 0$$

$$\text{From NEC } \rho + p \geq 0$$